PPO Lane Keeping — v14 Results

CARLA 0.9.16 · Town10HD_Opt · Stable-Baselines3 PPO · Northeastern Explorer

Final training reward	Final episode length	Evaluation episode	Evaluation reward	Cruise speed	Lane offset
621	**937**	**3000/3000**	**2859**	**41.1**	**0.09 m**
ep_rew_mean (from -38)	steps (from 53)	steps, no failure	0.95 per step	km/h, target 40	mean |offset|, max 1.04

Setup

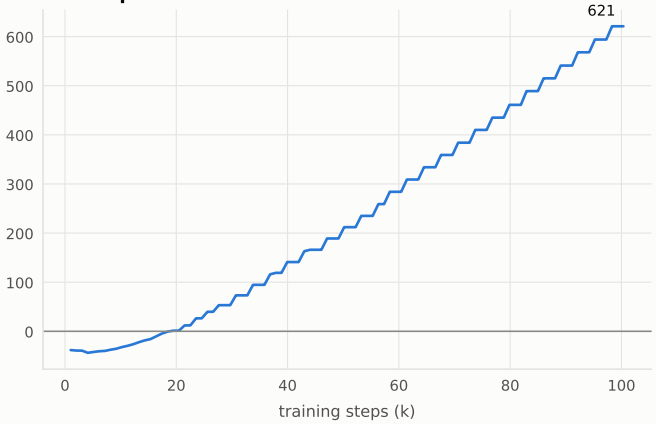
Algorithm	PPO (MlpPolicy), n_steps 1024, batch 64, lr 3e-4
Training	100,352 steps, 33 min, ~50 fps (CPU)
Normalization	VecNormalize (observations and reward)
Simulation	synchronous, 0.05 s per step (20 Hz)
Observation (8)	speed, offset, heading error, route heading 5/10/20/40 m, prev. steer
Action (2)	steer, throttle/brake (−0.3..0 coasts)
Reward	forward speed → 40 km/h − lane, heading, steer-change penalties
Termination	off-road, collision, stuck, > 6 m off route (−30)
Truncation	3000 steps or end of the 2000 m route

Notes

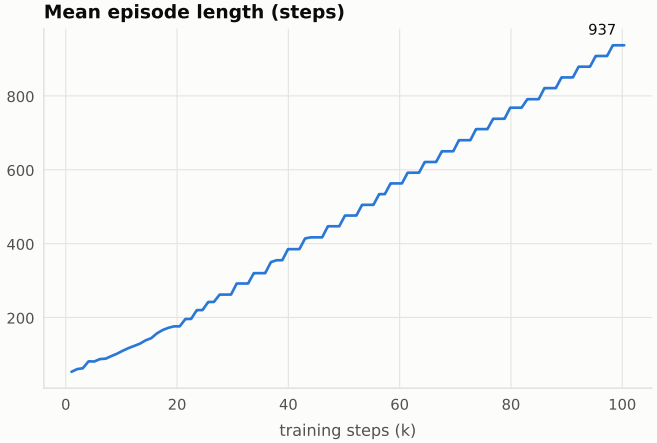
- Evaluation stopped at the time limit after 1,700 m; the last ~300 m of the route were not driven.
- Tested only on the training start point and route; generalization untested.
- Training ran on CPU: torch 2.14 rejected the node's CUDA 12.3 driver.
- Training episodes are stochastic (exploration) and end earlier than the deterministic evaluation.

Training progress

Mean episode reward

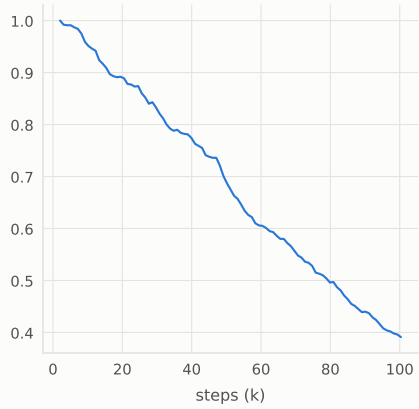


Mean episode length (steps)

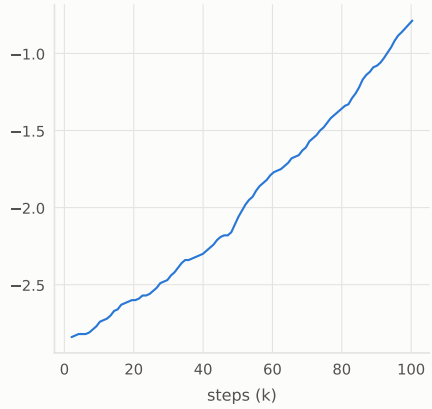


PPO diagnostics

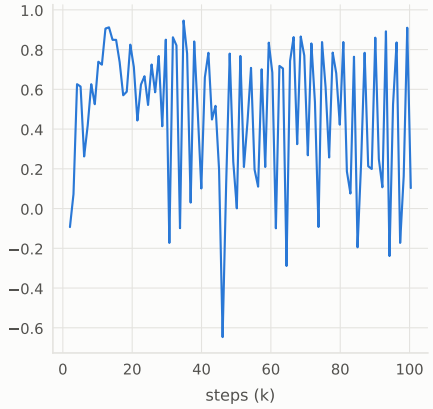
Policy std (exploration)



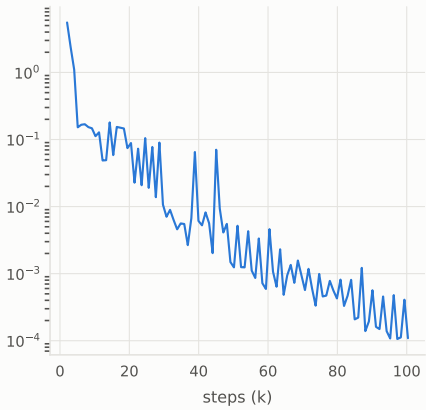
Entropy loss



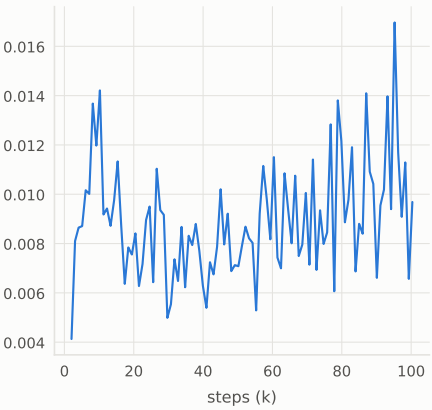
Explained variance



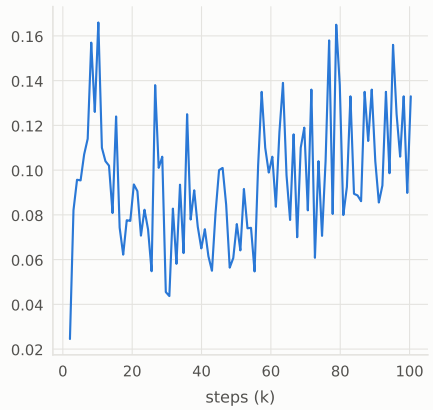
Value loss (log)



Approx. KL

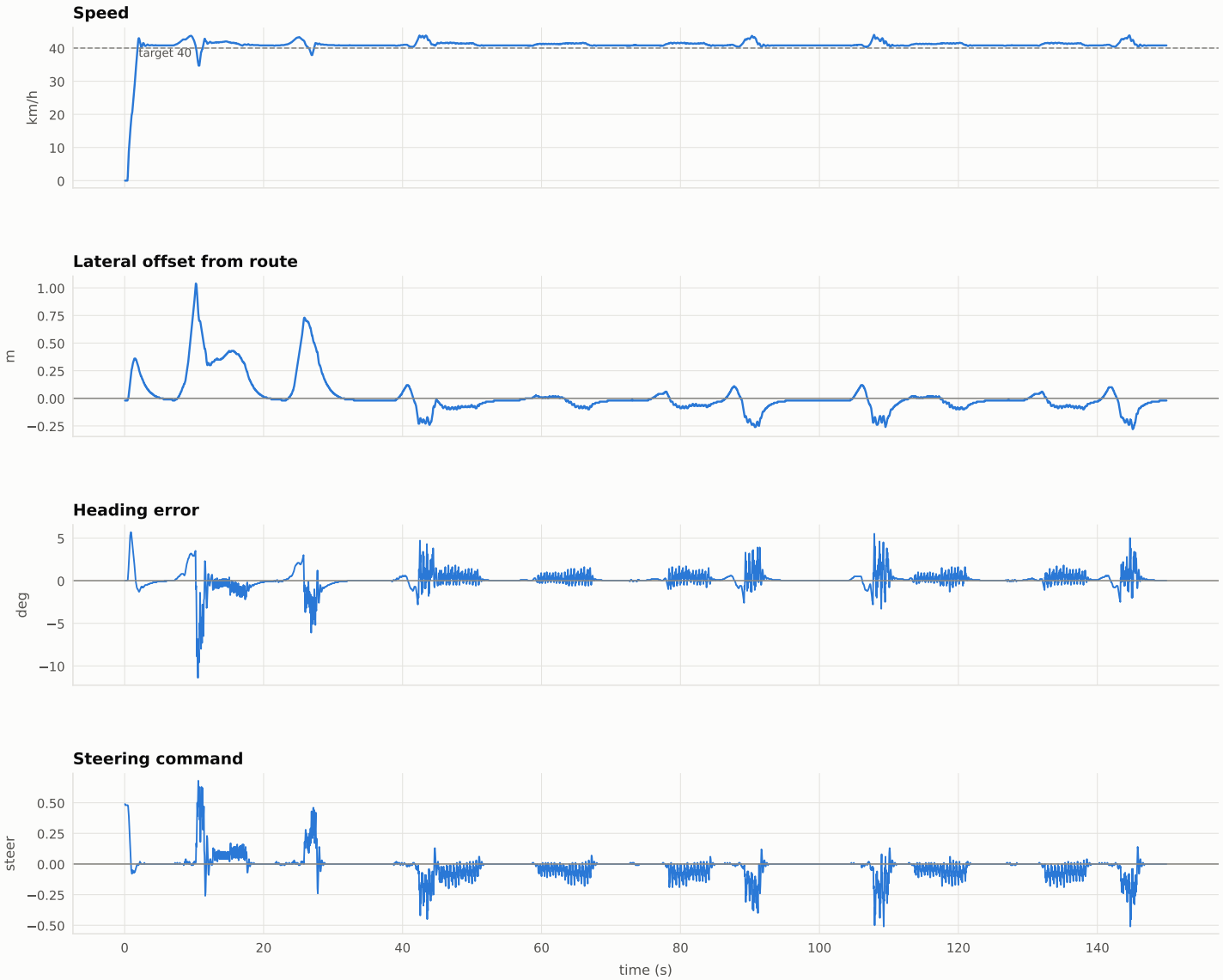


Clip fraction

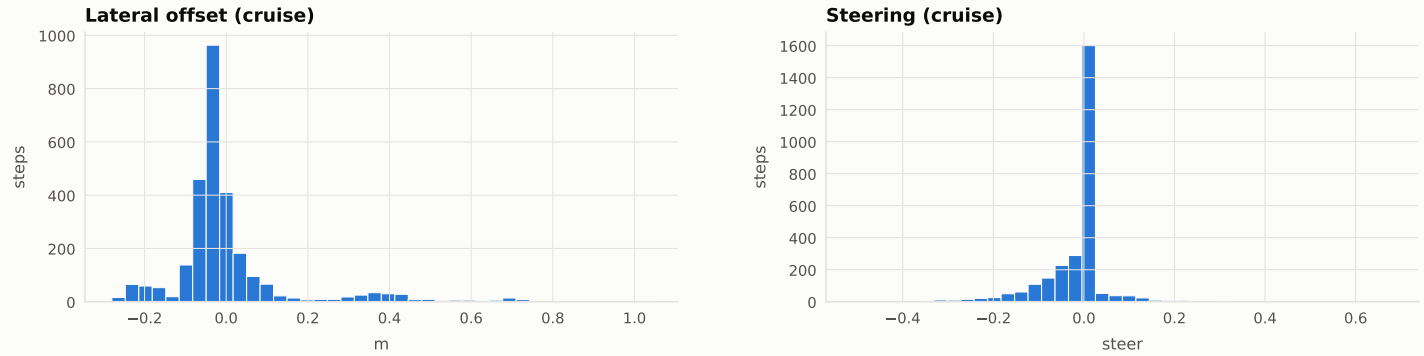


Evaluation run

Deterministic policy, one episode, 3000 steps (150 s). Statistics use the cruise phase (after step 200).



In curves steering and heading switch rapidly (dense bands): a residual zigzag the steer-change penalty does not fully remove.



	mean	std	min	max
Speed (km/h)	41.096	0.696	34.700	44.000
Offset (m)	0.007	0.158	-0.280	1.040
|Offset| (m)	0.086	0.132	0.000	1.040
Heading (°)	0.062	1.010	-11.400	5.500
Steer	-0.018	0.087	-0.510	0.680
|Δ steer| per step	0.027	0.043	0.000	0.360
Throttle	0.460	0.056	0.020	1.000
Reward per step	0.960	0.032	0.700	0.990

Episode

Reward 2859.11

Distance 1,700 m of 2000 m

End reason: time limit

0 → 42 km/h by step 200 (10 s)

Max |offset| 1.04 m (first 30 s)